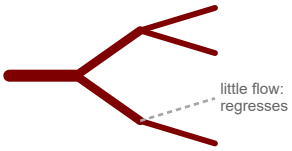
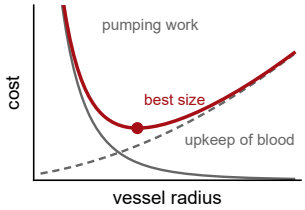
0 Where the rule comes from

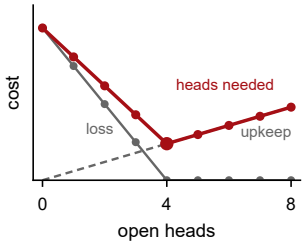
a Vessels are kept while they carry flow



b Murray's law: work + upkeep

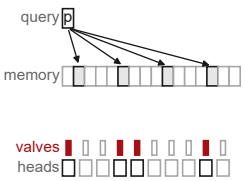


c Heads: loss + upkeep

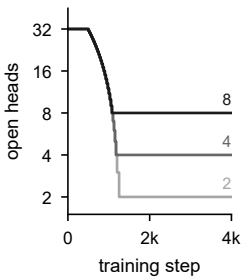


1 Does the rule keep exactly the heads a task needs?

d Retrieval task



e Heads closing

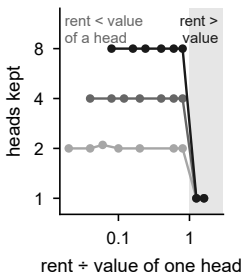


f Same 30 runs, every method

method	set by	exact	intact
Ours	upkeep	**29**	**30**
ZipLM	target	24	0
Michel et al.	target	28	0
L0 gates	penalty	9	0

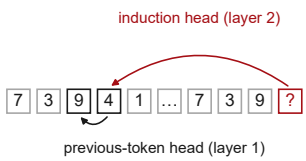
ZipLM uses our score, once

g Too much rent: collapse

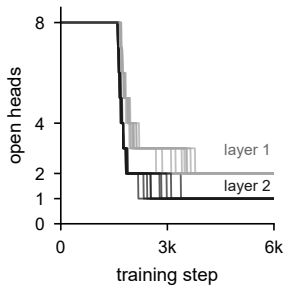


2 Does it find the known induction circuit?

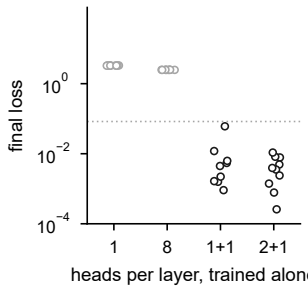
h Repeated random sequence, 2 layers × 8 heads



i Heads closing (10 runs)

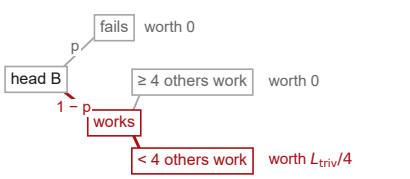


j 1+1 suffices: the 2nd layer-1 head is a backup



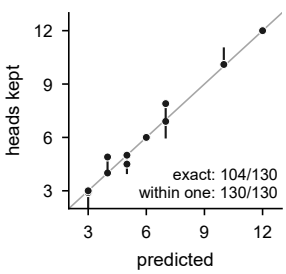
3 When is a backup head worth keeping?

k Is one more head worth its upkeep?

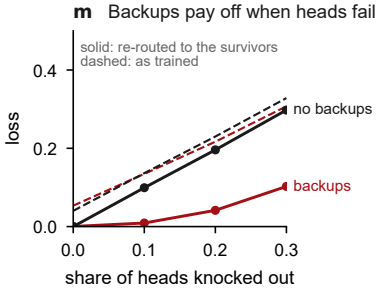


task needs 4 heads; keep head B while its value ≥ upkeep

l Prediction vs. runs



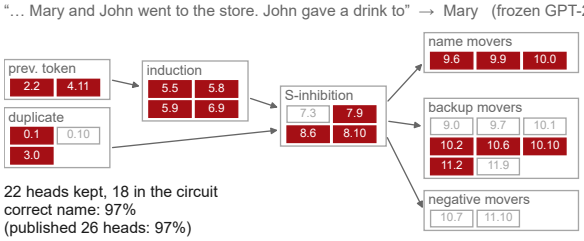
m Backups pay off when heads fail



4 Does it find a known circuit in a pretrained model?

n Published circuit; heads kept by the rule filled

"… Mary and John went to the store. John gave a drink to" → Mary (frozen GPT-2 small, 144 heads)



22 heads kept, 18 in the circuit
correct name: 97%
(published 26 heads: 97%)

o Same number of heads

